

$$Cl + e^- \rightarrow Cl^-$$

Figure. 5.4 Formation of chloride ion

Which atoms tend to lose electrons? Which are tend to gain electrons? Atoms that have 1, 2, 3 electrons in their valence shell tend to lose electrons whereas atoms having 5, 6, 7 valence electrons tend to gain electrons.

Table 13.2 Unstable electronic configuration

Element	Atomic number	Electron distribution	Valence shell electrons
Boron	5	2, 3	3
Nitrogen	7	2, 5	5
Oxygen	8	2,6	6
Sodium	11	2, 8, 1	1

13.2 Lewis dot structure

When atoms combine to form compounds, their valence electrons involve in bonding. Therefore, it is helpful to have a method to depict the valence electrons in the atoms. This can be done using Lewis dot symbol method. The Lewis dot structure or electron dot symbol for an atom consists of the symbol of the element surrounded by dots representing the electrons of the valence shell of the atom. The unpaired electron in the valence shell is represented by a single dot whereas the paired electrons are represented by a pair of dots.

Symbols other than dots, like crosses or circles may be used to differentiate the electrons of the different atoms in the molecule.

Table 13.3 Lewis dot structure

Element	Atomic number	Electron distribution	Valence shell electrons	Lewis dot structure
Hydrogen	1	1	1	H·
Helium	2	2	2	He:
Beryllium	4	2, 2	2	·Be·
Carbon	6	2, 4	4	·C·
Nitrogen	7	2, 5	5	·N·
Oxygen	8	2,6	6	·O:

More to Know

Note that dots are placed one to each side of the letter symbol until all four sides are occupied. Then the dots are written two to a side until all valence electrons are accounted for. The exact placement of the single dots is immaterial.

13.3 Types of chemical bond

All the elements have different valence shell electronic configuration. So the way in which they combine to form compounds also differs. Hence, there are different types of chemical bonding possible between atoms which make the molecules. Depending on the type of bond, they show different characteristics or properties. Such types of bonding, that are considered to exist in molecules, are categorized as shown Figure 13.5. Among these, let us learn about the Ionic bond, Covalent bond and Coordinate bond in this chapter and other types of bond in the higher classes.

13.3.1 Ionic (or) Electrovalent bond

An ionic bond is a chemical bond formed by the electrostatic attraction between

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